

Physics lab 4

Lucas Frykman

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Q1: Interference

Method: We simulate the time-dependent Schrödinger equation for a Gaussian wave packet reflecting off an infinite potential wall. We run the simulation at two different mesh resolutions ($N = 2^{13}$ and $N = 2^{14}$) to observe the resulting probability density as the incident and reflected waves overlap.

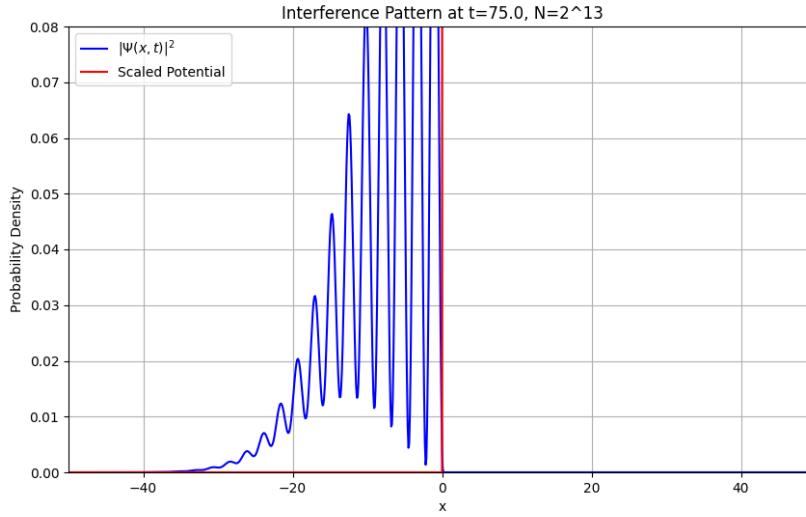


Figure 1: Interference Pattern ($N = 2^{13}$). Simulation parameters: $dx = 0.1, dt = 0.01, a = 8.0, x_0 = -100.0, E = 1.0, m = 1.0, \hbar = 1.0, t = 75.0$, with a potential wall $V_0 = 100.0$ for $x > 0$.

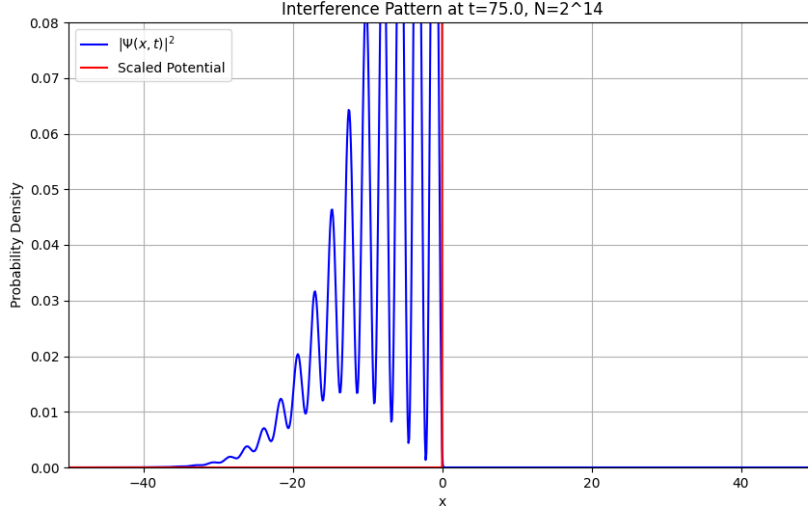


Figure 2: Interference Pattern ($N = 2^{14}$). Simulation parameters: $dx = 0.1, dt = 0.01, a = 8.0, x_0 = -100.0, E = 1.0, m = 1.0, \hbar = 1.0, t = 75.0$, with a potential wall $V_0 = 100.0$ for $x > 0$.

Results: The plots show a distinct interference pattern during the reflection. The pattern remains consistent across both mesh resolutions ($N = 2^{13}$ and $N = 2^{14}$), providing evidence that the observed interference pattern is real and doesn't depend on mesh resolution.

Q2: Gamow Tunneling

Method: We simulate the splitting of a wave packet colliding with the Gamow Coulomb barrier $V = \begin{cases} -V_0 & \text{for } x < R \\ C/x & \text{for } x > R \end{cases}$ where $C = R = 1$ and $V_0 = 0$ with $E = 0.9$. We then calculate the transmission probability T for several initial energies E and plot $\ln T$ against $1/\sqrt{E}$.

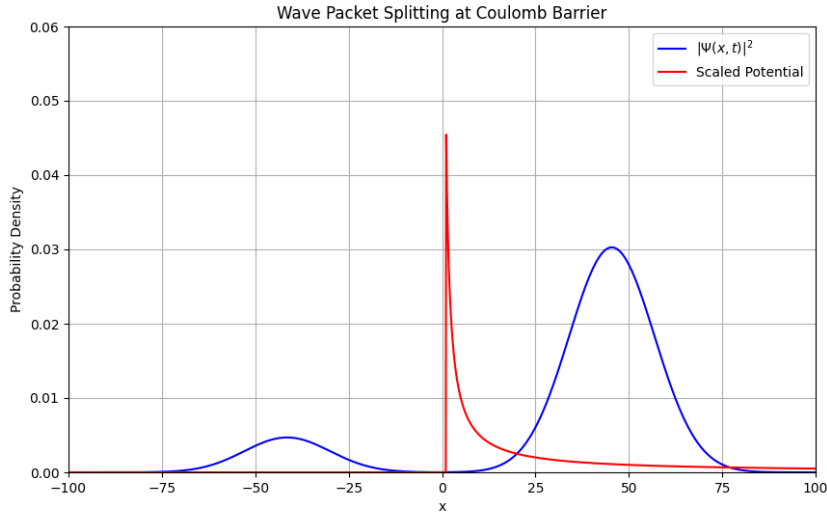


Figure 3: Wave Packet Splitting at Coulomb Barrier. Simulation parameters: $N = 2^{14}, dx = 0.1, dt = 0.01, a = 8.0, x_0 = -100.0, E = 0.9, m = 1.0, \hbar = 1.0, t = 110.0$, with Gamow potential $V(x) = 1.0/x$ for $x > 1.0$.

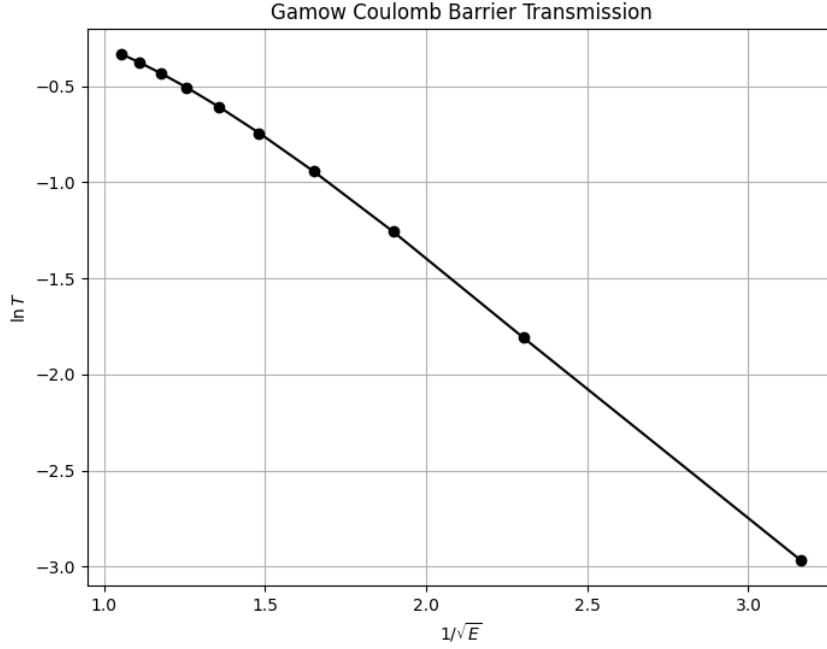


Figure 4: Transmission Probability T vs kinetic energy $E \in [0.1, 0.9]$. Simulation parameters: $N = 2^{14}$, $dx = 0.1$, $dt = 0.01$, $a = 8.0$, $x_0 = -100.0$, $m = 1.0$, $\hbar = 1.0$, with simulation times $t = 300.0/\sqrt{2E}$ and potential $V(x) = 1.0/x$ for $x > 1.0$.

Theory: According to Gamow’s theory of alpha decay, quantum tunneling through a Coulomb barrier predicts a transmission probability that depends exponentially on energy. Specifically, the theory states that $T \propto \exp(-k/\sqrt{E})$, which implies that a plot of $\ln T$ versus $1/\sqrt{E}$ should yield a straight line with a negative slope.

Results: In Figure 3, we observe that the wave packet splits into a reflected and transmitted component when encountering the Coulomb barrier, which is consistent with the tunneling effect predicted by quantum mechanics. We also observe that the transmitted wave packet is larger than the reflected one, which corresponds to a high transmission probability.

In Figure 4, the plot of $\ln T$ against $1/\sqrt{E}$ shows an approximately linear relationship at lower energies. This approximately linear region is consistent with Gamow’s predicted exponential dependence.

Q3: Fowler-Nordheim Tunneling

Method: We initialize a Gaussian wave packet with width $a = 10.0$, centered at $x_0 = -100.0$, and an initial energy $E = 0.5$. We utilize a spatial grid of $N = 2^{14}$ points with a step size $dx = 0.1$ and a time step $dt = 0.01$. We set the reduced Planck constant $\hbar = 1.0$ and the particle mass $m = 1.0$.

First, we simulate the wave packet encountering a flat potential step $V(x) = V_0 = 1.0$ for $x > 0$ to observe reflection. Then, we simulate field emission tunneling through a slanted potential barrier $V(x) = V_0(1 - x/L_{\text{barrier}})$ for $0 < x < L_{\text{barrier}}$ with $L_{\text{barrier}} = 4.0$. We calculate the transmission T past $x = 4.0$ for varying barrier heights $V_0 \in [0.6, 1.0]$ and plot $\ln T$ against $V_0^{3/2}$.

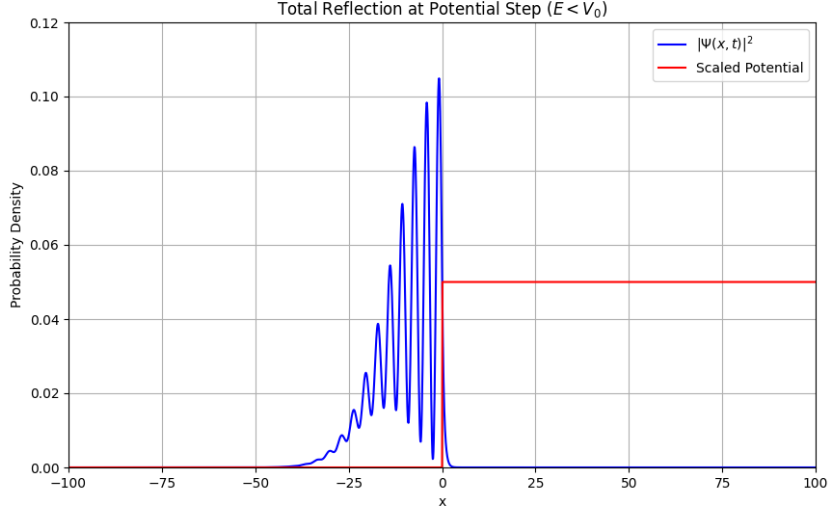


Figure 5: Complete Reflection at Flat Potential Step. Simulation parameters: $N = 2^{14}$, $dx = 0.1$, $dt = 0.01$, $a = 10.0$, $x_0 = -100.0$, $E = 0.5$, $m = 1.0$, $\hbar = 1.0$, $t = 110.0$, with a flat step potential $V_0 = 1.0$ for $x > 0$.

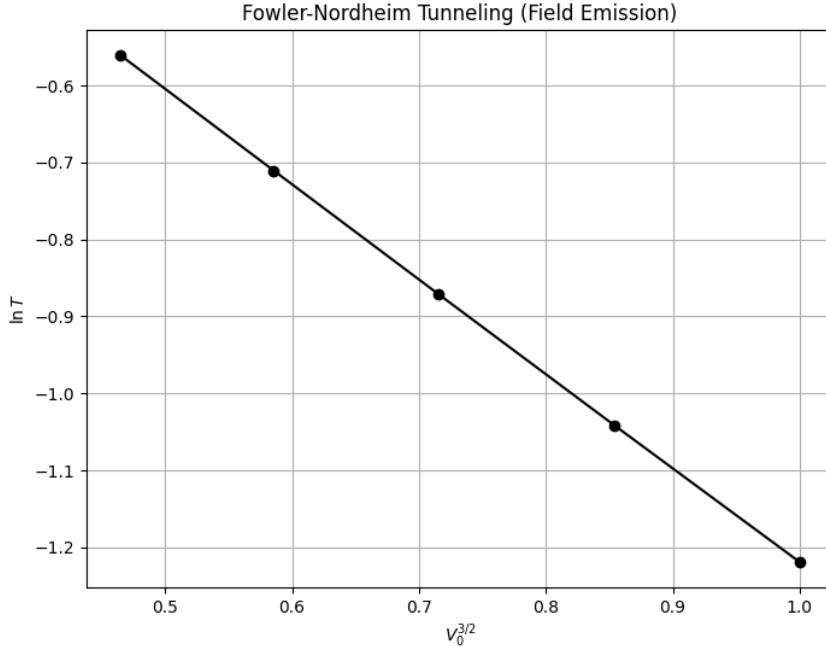


Figure 6: Fowler-Nordheim Tunneling. Simulation parameters: $N = 2^{14}$, $dx = 0.1$, $dt = 0.01$, $a = 10.0$, $x_0 = -100.0$, $E = 0.5$, $m = 1.0$, $\hbar = 1.0$, with a slanted potential barrier $V(x) = V_0(1 - x/4.0)$ for $0 < x < 4.0$ and varying $V_0 \in [0.6, 1.0]$.

Theory: According to the Fowler-Nordheim theory of field emission, the transmission probability through a triangular potential barrier is given by $T \propto \exp(-c \cdot V_0^{3/2}/F)$. Therefore, we expect a plot of $\ln T$ versus $V_0^{3/2}$ to yield a straight line.

Results: Figure 5 shows negligible transmission and predominantly reflected probability density, as expected for a potential step with $E < V_0$. Figure 6, the plot of $\ln T$ against $V_0^{3/2}$ for the slanted potential, shows an approximately linear relationship, which is consistent with the exponential dependence

predicted by Fowler-Nordheim theory.